# Data Analysis & Probability

ClipSAT Printable Practice Suite (clipsat.org) • Topic 7.01

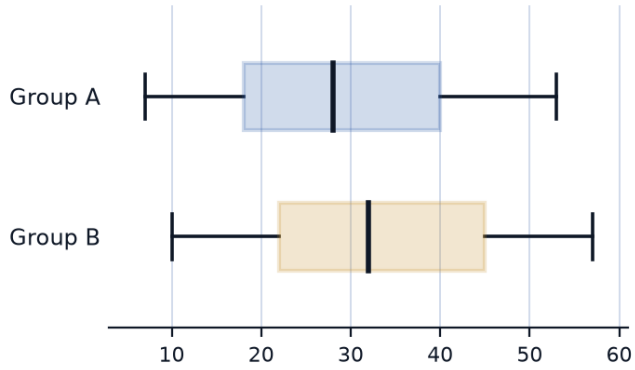
## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** [Foundational] • Target: Mean, median, mode and range

Find the mean of $6, 9, 12, 15, 18$.

**2** [Foundational] • Target: Mean, median, mode and range

Ages of 7 students: $13, 15, 13, 16, 14, 13, 17$. Find the mean, median, and mode.




**3** [Foundational] • Target: Mean, median, mode and range

Find the range and mode of the data set $8, 3, 8, 15, 8, 22, 3$.

**4** [Foundational] • Target: Mean, median, mode and range

A data set has mean $18$ and 6 values. Find the sum of the values.

**5** [Foundational] • Target: Mean, median, mode and range

The data set $10, 12, 11, 13, 50$ contains an outlier. Find the mean before and after removing $50$.

**6** [Foundational] • Target: Mean, median, mode and range

A data set of 5 values has mean $20$. Four of the values are $15, 18, 22, 25$. Find the fifth value.

**7** [Foundational] • Target: Reading tables

A survey of 50 people's favorite fruit: Apple 15, Banana 12, Orange 10, Mango 13. What fraction of people chose banana?

## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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**8** [Intermediate] • Target: Reading tables

Using the same survey (Apple 15, Banana 12, Orange 10, Mango 13, out of 50), what percent chose either apple or mango?

**9** [Intermediate] • Target: Reading tables

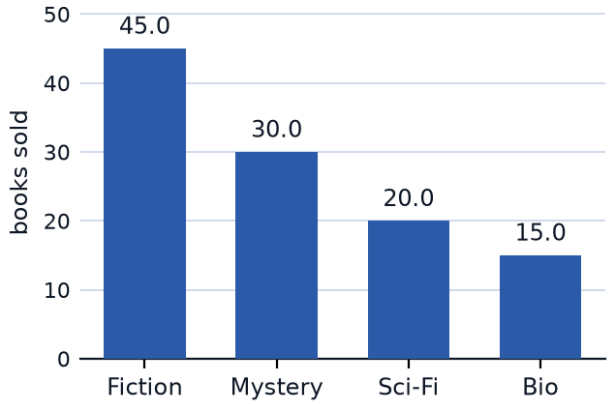
A table shows quiz scores: score 5 (3 students), score 7 (5), score 8 (8), score 9 (4), score 10 (2). Find the mode score and the total number of students.

**10** [Intermediate] • Target: Reading tables

Using the same quiz data, find the mean score to two decimal places.

**11** [Intermediate] • Target: Reading bar charts

The bar chart shows books sold by genre in one week. Find the total books sold and the modal (most common) genre.

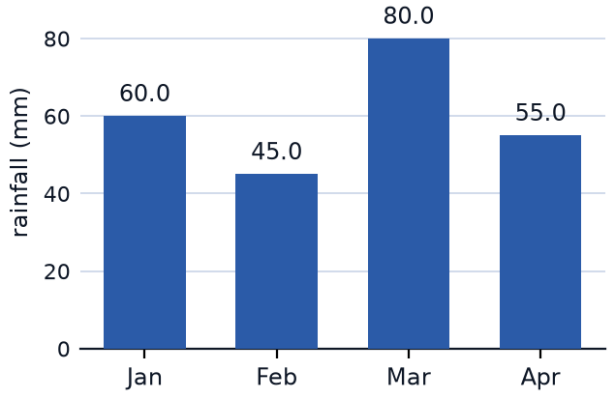



**12** [Intermediate] • Target: Reading bar charts

Using the same book-sales data, what percent of total books sold were mystery novels, to the nearest percent?

**13** [Intermediate] • Target: Reading bar charts

The bar chart shows monthly rainfall in mm. Find the mean monthly rainfall.




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**14** [Intermediate] • Target: Reading bar charts

Using the same rainfall data (Jan 60, Feb 45, Mar 80, Apr 55 mm), find the percent decrease from March to April.

## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

**15** [Challenging] • Target: Basic probability • Cross-Lesson: 8.01 Word Problems & Reasoning

A bag holds 4 red and 6 blue marbles. Find $P(\text{red})$.

**16** [Challenging] • Target: Basic probability • Cross-Lesson: 1.01 Arithmetic & Number Sense

A fair six-sided die is rolled. Find $P(\text{a number greater than 4})$.

**17** [Challenging] • Target: Basic probability • Cross-Lesson: 2.01 Ratios, Proportions & Percentages

A card is drawn from a standard deck of 52 cards. Find $P(\text{a heart})$.

**18** [Challenging] • Target: Basic probability • Cross-Lesson: 3.01 Algebra Essentials

A spinner has 8 equal sections numbered 1 to 8. Find $P(\text{an even number})$.

**19** [Challenging] • Target: Combined and complementary events • Cross-Lesson: 4.01 Geometry & Measurement

A bag holds 3 red, 5 blue and 2 green marbles. Find $P(\text{not red})$.

**20** [Challenging] • Target: Combined and complementary events • Cross-Lesson: 5.01 Quantitative Comparisons

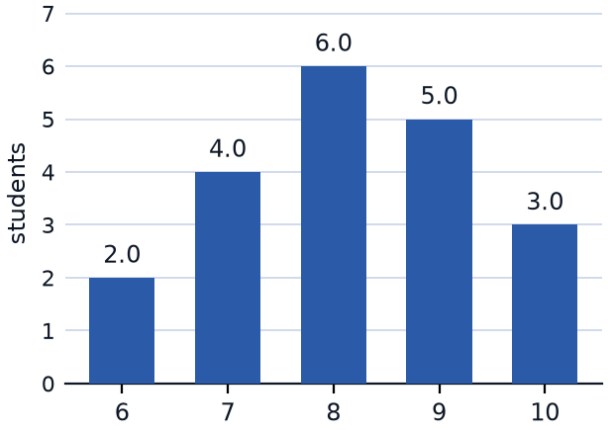
Two fair coins are tossed. Find $P(\text{both land heads})$.

## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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**21** [Foundational] • Target: Mean, median, mode and range

The bar chart shows quiz scores. Find the total number of students, and the modal score.




**a.** 20 students; mode $= 8$

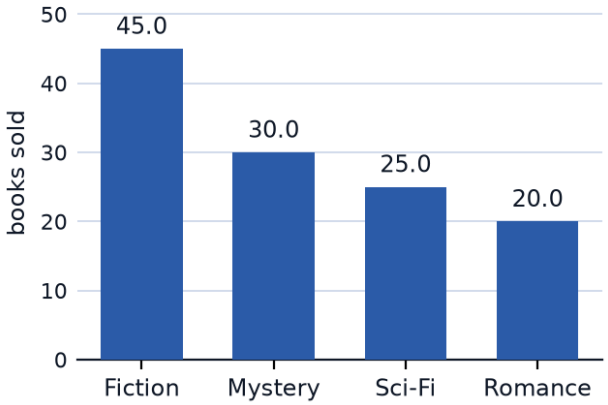
**b.** 20 students; mode $= 6$

**c.** 20 students; mode $= 9$

**d.** 18 students; mode $= 8$

**22** [Foundational] • Target: Reading bar charts

The bar chart shows books sold by genre in one week. Find the total books sold, and the modal genre.




**a.** 120; Romance

**b.** 100; Fiction

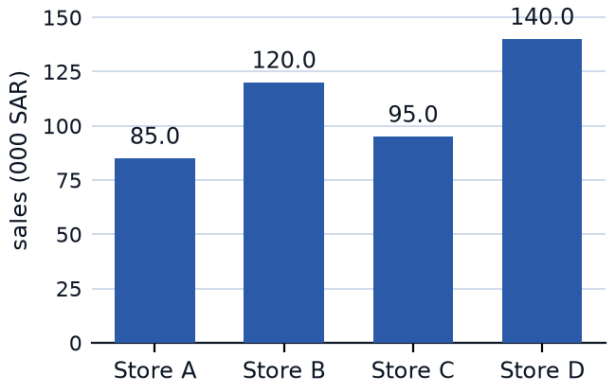
**c.** 120; Mystery

**d.** 120; Fiction

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**23** [Foundational] • Target: Reading bar charts

The bar chart shows quarterly sales for four stores, in thousands of riyals. Find the range, and which store had the highest sales.




**a.** 55; Store B

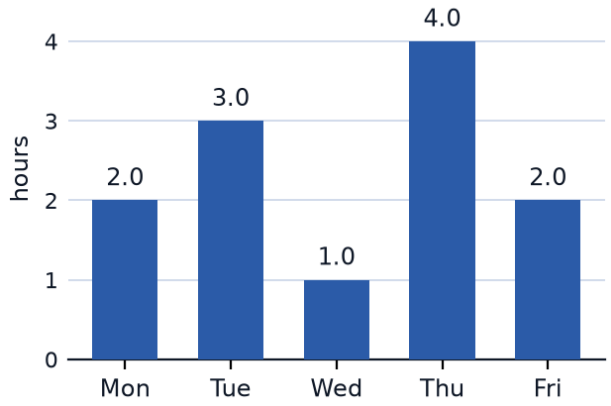
**b.** 55; Store D

**c.** 140; Store D

**d.** 35; Store D

**24** [Foundational] • Target: Reading bar charts

The bar chart shows hours studied each weekday. Find the total hours, and the mean hours per day.




**a.** 12; 2

**b.** 12; 2.4

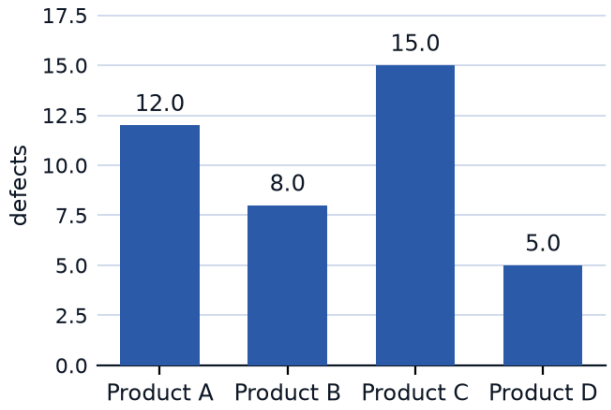
**c.** 12; 3

**d.** 10; 2

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**25** [Foundational] • Target: Reading bar charts

The bar chart shows the number of defective items found for four products in a quality check. Find the total defects, and the range.




**a.** 40; 10

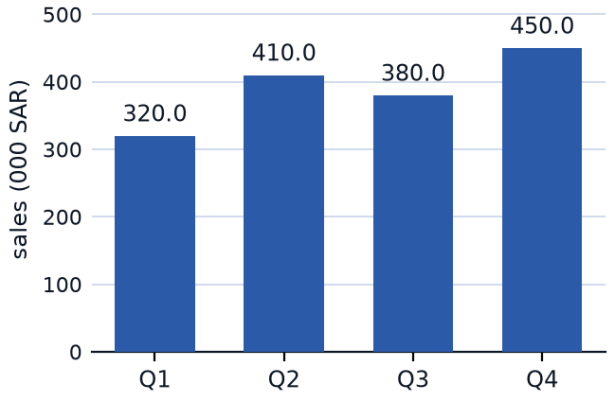
**b.** 40; 7

**c.** 35; 10

**d.** 40; 15

**26** [Foundational] • Target: Word problems

The bar chart shows a company's quarterly sales, in thousands of riyals. Find the total annual sales, and the mean quarterly sales.




**a.** 1,560; 450

**b.** 1,560; 390

**c.** 1,560; 400

**d.** 1,200; 390

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**27** [Foundational] • Target: Mean, median, mode and range

Ages of 6 students: $12, 14, 12, 15, 13, 12$. Find the mean, median, and mode.

**a.** Mean $= 13$, median $= 12$, mode $= 12.5$

**b.** Mean $= 13$, median $= 12.5$, mode $= 12$

**c.** Mean $= 12.5$, median $= 13$, mode $= 12$

**d.** Mean $= 13$, median $= 12.5$, mode $= 13$

**28** [Foundational] • Target: Mean, median, mode and range

A data set of 6 values has mean $25$. Five of the values are $20, 22, 28, 30, 32$. Find the sixth value.

**a.** 18

**b.** 28

**c.** 8

**d.** 150

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** [Intermediate] • Target: Reading tables

A survey of 60 people's favorite drink: Tea 18, Coffee 22, Juice 12, Water 8. What fraction of people chose coffee?

**a.** $\frac{18}{60}$

**b.** $\frac{22}{60}$

**c.** $\frac{11}{30}$

**d.** $\frac{11}{60}$

**30** [Intermediate] • Target: Reading tables

Using the same survey (Tea 18, Coffee 22, Juice 12, Water 8, out of 60), what percent chose either tea or water, to the nearest percent?

**a.** 43%

**b.** 30%

**c.** 50%

**d.** 26%

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**31** [Intermediate] • Target: Reading tables

A table shows test scores: score 60 (4 students), 70 (7), 80 (9), 90 (6), 100 (4). Find the total number of students.

**a.** 34

**b.** 30

**c.** 26

**d.** 25

**32** [Intermediate] • Target: Reading tables

Using the same test data, find the mean score to two decimal places.

**a.** 79.67

**b.** 80.00

**c.** 76.33

**d.** 79.00

**33** [Intermediate] • Target: Reading tables

A table shows the number of pets owned by 25 families: 0 pets (6 families), 1 pet (10), 2 pets (7), 3 pets (2). Find the median number of pets.

**a.** 0

**b.** 1.2

**c.** 2

**d.** 1

**34** [Intermediate] • Target: Reading tables

Using the same pet data, find the mean number of pets per family, to two decimal places.

**a.** 1.00

**b.** 1.20

**c.** 1.50

**d.** 30

**35** [Intermediate] • Target: Reading bar charts

Using the same rainfall data (Jan 50, Feb 70, Mar 40, Apr 65 mm), find the percent decrease from Feb to Mar, to the nearest percent.

**a.** 75%

**b.** 43%

**c.** 30%

**d.** 40%

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**36** [Intermediate] • Target: Basic probability

A fair six-sided die is rolled. Find $P(\text{a number less than } 3)$.

**a.** $\frac{1}{6}$

**b.** $\frac{1}{2}$

**c.** $\frac{2}{3}$

**d.** $\frac{1}{3}$

**37** [Intermediate] • Target: Basic probability

A card is drawn from a standard deck of 52 cards. Find $P(\text{a king})$.

**a.** $\frac{1}{4}$

**b.** $\frac{1}{52}$

**c.** $\frac{1}{13}$

**d.** $\frac{4}{13}$

## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • Target: Basic probability • Cross-Lesson: 3.01 Algebra Essentials

A spinner has 10 equal sections numbered 1 to 10. Find $P(\text{a multiple of } 3)$.

**a.** $\frac{1}{10}$

**b.** $\frac{1}{3}$

**c.** $\frac{4}{10}$

**d.** $\frac{3}{10}$

**39** [Challenging] • Target: Basic probability • Cross-Lesson: 4.01 Geometry & Measurement

A bag holds 6 yellow and 9 green balls. Find $P(\text{yellow})$.

**a.** $\frac{1}{5}$

**b.** $\frac{6}{9}$

**c.** $\frac{3}{5}$

**d.** $\frac{2}{5}$

**40** [Challenging] • Target: Basic probability • Cross-Lesson: 5.01 Quantitative Comparisons

A fair six-sided die is rolled. Find $P(\text{an odd number})$.

**a.** $\frac{2}{3}$

**b.** $\frac{1}{6}$

**c.** $\frac{1}{2}$

**d.** $\frac{1}{3}$

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**41** [Challenging] • Target: Combined and complementary events • Cross-Lesson: 6.01 Sequences & Patterns

A bag holds 4 red, 6 blue and 5 green marbles. Find $P(\text{not blue})$.

**a.** $\frac{3}{5}$

**b.** $\frac{9}{6}$

**c.** $\frac{2}{5}$

**d.** $\frac{6}{15}$

**42** [Challenging] • Target: Combined and complementary events • Cross-Lesson: 8.01 Word Problems & Reasoning

Two fair coins are tossed. Find $P(\text{at least one tail})$.

**a.** $\frac{3}{4}$

**b.** $\frac{1}{3}$

**c.** $\frac{1}{4}$

**d.** $\frac{1}{2}$

**43** [Challenging] • Target: Combined and complementary events • Cross-Lesson: 1.01 Arithmetic & Number Sense

A die is rolled twice. Find $P(\text{both rolls show an even number})$.

**a.** $\frac{1}{6}$

**b.** $\frac{1}{3}$

**c.** $\frac{1}{4}$

**d.** $\frac{1}{2}$

**44** [Challenging] • Target: Combined and complementary events • Cross-Lesson: 2.01 Ratios, Proportions & Percentages

A bag holds 5 white and 7 black balls. One ball is drawn and not replaced, then a second is drawn. Find $P(\text{both black})$.

**a.** $\frac{7}{22}$

**b.** $\frac{6}{11}$

**c.** $\frac{49}{144}$

**d.** $\frac{7}{12}$

**45** [Challenging] • Target: Combined and complementary events • Cross-Lesson: 3.01 Algebra Essentials

A card is drawn, replaced, then a second card is drawn from a standard deck. Find $P(\text{both hearts})$.

**a.** $\frac{1}{8}$

**b.** $\frac{1}{2}$

**c.** $\frac{1}{4}$

**d.** $\frac{1}{16}$

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